`bispectrum`: Selective G-Bispectra Made Practical

Anonymous Author(s)
Affiliation
Address
email

Abstract

Many machine-learning tasks are invariant under group actions: image classification under planar rotation $\mathrm{SO}(2)$, signal classification under translation, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. The G-bispectrum is a principled *complete* invariant (it preserves all signal information up to the group action) but its deployment has been hampered by quadratic cost and a patchwork of group-specific implementations. We present `bispectrum`, an open-source, fully unit-tested PyTorch library that computes *selective* G-bispectral invariants for seven different group/domain pairs as drop-in differentiable `nn.Modules`. For finite groups, selectivity reduces the invariant size from $O(|G|^2)$ for the full bispectrum to $O(|G|)$ while remaining complete on generic signals. For $\mathrm{SO}(3)$ on S^2, we introduce an augmented selective construction Φ_{sel} with $\Theta(L^2)$ coefficients at band-limit L (matching the orbit-space lower bound $(L+1)^2 - 3$ up to a constant, vs. $O(L^3)$ for the full bispectrum); a formal completeness proof for the continuous case is left to future work. Gradient-descent reconstruction from Φ_{sel} produces visually aligned reconstructions with relative image-space residuals in the 0.1–0.3 range across our digit sweep (Section 6.2). We benchmark the library on three datasets across three group/domain pairs — PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$ — comparing against norm pooling, gated pooling, Fourier-ELU pooling, max pooling, and (non-equivariant) augmented CNN baselines. The library delivers near-exact rotation invariance (test-set rotation-accuracy std $\sigma_{\mathrm{rot}} < 10^{-3}$) and a data-efficiency advantage over incomplete alternatives in the low-data, moderate-capacity regime; the selective forward pass runs in sub-millisecond time on GPU for finite groups. Code is available at `https://anonymous.4open.science/r/bispectrum/`.

1 Introduction

Many machine-learning tasks are invariant under the action of a symmetry group G: histopathology classification under planar rotations C_n, signal classification under translations, volumetric organ classification under cube rotations O, and spherical-image classification under 3D rotations $\mathrm{SO}(3)$. Equivariant neural networks [Cohen and Welling, 2016, Weiler and Cesa, 2019] enforce these symmetries by construction in their feature maps, but at the point where a prediction must be made any task that does not depend on the group action — classification, regression, detection — still has to map the equivariant features to an *invariant* representation.

The *G-bispectrum* is a complex-valued vector of triple-correlation coefficients of the group Fourier transform. It has three properties that matter for ML: (i) **completeness** — it preserves all signal information up to the group action (the map is one-to-one on group orbits); (ii) **inversion** — the signal can in principle be recovered from the invariant up to its orbit; and (iii) **selectivity** — a small $O(|G|)$ index subset of the $O(|G|^2)$ full bispectrum still suffices for completeness on generic signals [Mataigne et al., 2024].

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Existing invariant pooling layers — norm pooling, gate-based pooling, max/sum aggregation [Weiler and Cesa, 2019] — are simple but *incomplete*: they discard information distinguishing signals in different orbits. The bispectrum is a principled alternative [Giannakis, 1989, Kakarala, 2012, Sanborn et al., 2023, Mataigne et al., 2024], but existing constructions are scattered across theory papers and domain-specific implementations, and the selective template of Mataigne et al. [2024] is specific to finite groups; the continuous case of $\mathrm{SO}(3)$ on S^2 has remained open.

This paper presents `bispectrum`, a PyTorch [Paszke et al., 2019] library for computing bispectral invariants (and, where available, their inverses) on the C_n, D_n, $\mathbb{T}^d$, disk, S^2, and octahedral domains as drop-in `nn.Modules`.

Contributions.

1. **Library.** PyTorch `nn.Modules` for seven G/domain pairs with a uniform API; selective $O(|G|)$ default, precomputed CG buffers, GPU-optimized forward passes (<1 ms for finite groups), 96.8% test coverage.
2. **Theory.** A selective $\mathrm{SO}(3)$ invariant on S^2, Φ_{sel}, with $\Theta(L^2)$ coefficients (Sec. 6), matching the orbit-space lower bound $(L+1)^2 - 3$; reconstruction evidence supports that it preserves substantial orbit information, but a formal completeness proof is left to future work.
3. **Benchmarks.** PCam under C_8, OrganMNIST3D under the octahedral group, and Spherical MNIST under $\mathrm{SO}(3)$, against norm / gated / Fourier-ELU / max pooling and augmented CNN baselines; near-exact invariance and a data-efficiency advantage over incomplete alternatives at low data.

2 Mathematical Background

Notation. For a group G with associative product, a *representation* ρ assigns each $g \in G$ a unitary matrix $\rho(g)$ with $\rho(gh) = \rho(g)\rho(h)$; an *irreducible representation* (*irrep*) has no nontrivial invariant subspace, and the *dual* $\hat{G}$ is the set of equivalence classes of irreps (for finite G, $|\hat{G}| = O(|G|)$). A G-action on a domain X, $(g, x) \mapsto g \cdot x$, induces $(f \circ g)(x) := f(g \cdot x)$; two signals are in the same *orbit* iff related by some $g \in G$, and an *invariant* of f is a function constant on each orbit. For compact G, $d\mu$ denotes the normalised Haar measure (counting measure for finite G). "Lebesgue-almost-every f" means outside a measure-zero algebraic subvariety. The groups and domains we target are $C_n, D_n, \mathbb{T}^d, \mathrm{SO}(2)$ on the disk, $\mathrm{SO}(3)$ on S^2, and the chiral octahedral group O on 3D voxel grids.

2.1 Equivariant features and invariant maps

A neural network layer Φ is *G-equivariant* if its output transforms predictably under the group action: $[\pi_{\mathrm{out}}(g) \circ \Phi](f) = \Phi(\pi_{\mathrm{in}}(g)f)$ for all $g \in G$ [Cohen and Welling, 2016, Weiler and Cesa, 2019]. Here f is an input signal (a function on the domain X), $\pi_{\mathrm{in}}, \pi_{\mathrm{out}}$ are the (representations of the) G-actions on the input and output feature spaces of Φ, and the composition $\pi_{\mathrm{out}}(g) \circ \Phi$ is the action of g on the output side. Stacking equivariant layers preserves equivariance, but any task that requires a prediction independent of the group action—classification, regression, detection—must eventually map equivariant features to *invariant* ones (constant on each orbit).

Table 1 summarises the trade-offs between exactness, completeness, and computational cost across existing invariant pooling layers. Norm pooling and gated nonlinearities are exact invariants but *incomplete*: they discard the phase of Fourier blocks, collapsing distinct orbits. Fourier-space pointwise nonlinearities preserve completeness but sacrifice exactness through aliasing [Weiler and Cesa, 2019]. The full bispectrum is exact and complete but costs $O(|G|^2)$ coefficients. The *selective* bispectrum (introduced below) preserves both exactness and completeness while using only $O(|G|)$ coefficients on generic signals.

Table 1: Invariant nonlinearities for equivariant CNNs. "Exact"* means the invariant has no discretisation/aliasing error; "Complete" means it separates orbits; "Genericity assumption"† means completeness holds outside a Lebesgue-measure-zero algebraic subvariety. The selective bispectrum is the only entry that is simultaneously exact, complete, and linear-cost for finite groups; the $\mathrm{SO}(3)$-on-S^2 case is treated empirically (Sec. 6).

Nonlinearity	Exact?*	Complete?	Coefficients	Forward cost	Key limitation
Norm	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Discards phase
Gate	Yes	No	$O(\|G\|)$	$O(\|G\|)$	Scalar bottleneck
Fourier pointwise	No	Yes‡	$O(\|G\|)$	$O(\|G\| \log \|G\|)$	Aliasing
Full bispectrum	Yes	Yes	$O(\|G\|^2)$	$O(\|G\|^3)$	Quadratic cost
Selective bispectrum (finite G)	**Yes**	**Yes**	$\mathbf{O(\|G\|)}$	$\mathbf{O(\|G\|^2)}$	Genericity assumption†
Augmented selective $\mathrm{SO}(3)/S^2$ (ours)	Yes	Empirical	$\Theta(L^2)$	$O(L^3)$	Proof of concept only

‡Complete pre-aliasing; the pointwise nonlinearity introduces aliasing that may break completeness in practice.

2.2 The G-bispectrum and selective bispectra

Let G be a compact group with dual $\hat{G}$ and irreps $\{\rho_i\}_{i\in\hat{G}}$ of dimensions d_i. The *group Fourier transform* [Diaconis and Rockmore, 1990] of a signal $f : G \to \mathbb{C}$ is

$$\hat{f}(\rho_i) = \int_G f(g)\, \rho_i(g)\, d\mu(g) \quad \in \mathbb{C}^{d_i \times d_i}, \tag{1}$$

which specialises to $\sum_{g\in G} f(g)\, \rho_i(g)$ when G is finite (counting measure). The Clebsch–Gordan (CG) decomposition $\rho_i \otimes \rho_j \cong \bigoplus_{k\in\hat{G}} m_{ij}^k \rho_k$ expresses the tensor product of two irreps as a direct sum of irreps (with integer multiplicities $m_{ij}^k \geq 0$); the change-of-basis is realised by the *CG isometry* $C_{ij}^k : V_{\rho_k} \to V_{\rho_i} \otimes V_{\rho_j}$, projecting the tensor product onto its ρ_k-isotypic component. For each admissible triple (i, j, k) with $m_{ij}^k > 0$, the *bispectrum* is the scalar

$$\beta_{i,j}^k(f) := \big\langle\, (C_{ij}^k)^\dagger \big(\hat{f}(\rho_i) \otimes \hat{f}(\rho_j)\big),\, \hat{f}(\rho_k)\big\rangle. \tag{2}$$

Each $\beta_{i,j}^k$ is invariant under the action $f \mapsto f \circ g$. The full matrix/operator bispectrum is complete on compact groups under Kakarala-style hypotheses [Kakarala, 1992, 2012]; for finite G, the scalar-entry family $\{\beta_{i,j}^k\}_{(i,j,k)}$ remains complete on generic signals via the selective construction of Mataigne et al. [2024]. For $\mathrm{SO}(3)$ on S^2 we do not claim scalar completeness—the library's selective invariant is empirical; see Section 6. For finite G, $|\hat{G}| = O(|G|)$, so the full bispectrum has $O(|G|^2)$ scalar coefficients (indexed by $(i, j) \in \hat{G}^2$ with k ranging over the CG-allowed irreps). Equation (2) is stated for G acting on itself; for homogeneous spaces of the form G/H (e.g. $\mathrm{SO}(3)$ on S^2, or $\mathrm{SO}(2)$ on the disk), the same construction applies after expanding f in a basis adapted to G/H (spherical harmonics for S^2, Bessel modes on the disk). The explicit specialisation for S^2 is in Definition 3.

Selective bispectra. Mataigne et al. [2024] show that for finite groups, a breadth-first search (BFS) over the Kronecker product table of $\hat{G}$ selects $O(|G|)$ bispectral coefficients $\beta_{i,j}^k$ that remain complete; Myers and Miolane [2025] extend this to the disk. This reduction from quadratic to linear is the key to practical deployment: for example, the bispectrum of the octahedral group drops from 576 to 172 coefficients.

3 Related Work

Equivariant network libraries. `escnn` [Cesa et al., 2022] and `e3nn` [Geiger and Smidt, 2022] provide general-purpose equivariant convolution layers for 2D and 3D groups respectively, and `lie-nn` [Batatia et al., 2023] provides general Lie-group representation primitives (irreducible representations, Clebsch–Gordan decompositions, tensor products) on top of which custom equivariant layers can be built. These libraries implement equivariant *feature maps* but leave the choice of invariant pooling to the user (typically norm or gate). Our library is complementary: it provides the invariant map as a drop-in `nn.Module` that plugs into architectures built with any of these frameworks.

Table 2: Library profiling summary on a single NVIDIA H100 80 GB GPU. "Sel. coefs" = selective coefficient count; "Fwd" /"Inv" = median forward/inversion wall-clock (ms); "–" = not implemented. *`SO3onS2` at $\ell_{\max}$=16. The full bispectrum at matched group order is 8256/524,800 coefficients for C_{128}/$\mathbb{T}^2$ (vs. 128/1024 selective, a 64×/512× reduction), taking 9.17/0.71 ms in the forward pass.

Module	G	$\lvert G\rvert$	Sel. coefs	Fwd (ms)	Inv (ms)
`CnonCn`	C_{128}	128	128	0.15	8.56
`TorusOnTorus`	$C_{32}\times C_{32}$	1024	1024	0.08	68.02
`DnonDn`	D_{32}	64	245	2.52	42.33
`SO2onDisk`	$\mathrm{SO}(2)$	∞	411	1.94	17.69
`SO3onS2`*	$\mathrm{SO}(3)$	∞	430	26.54	–
`OctaonOcta`	O	24	172	0.70	–

Bispectral approaches in neural networks. Sanborn et al. [2023] introduced bispectral neural networks (BNNs), using the full G-bispectrum as an invariant nonlinearity for C_n and D_n. Oreiller et al. [2022] applied bispectral CNNs to medical imaging with a fixed set of hand-selected triples. Both use the full (quadratic-cost) bispectrum and provide implementations tied to specific architectures. Our library generalizes to seven group/domain pairs, defaults to the $O(|G|)$ selective reduction [Mataigne et al., 2024], and exposes a framework-agnostic API. Chevalley et al. [2024] explore bispectral signatures for data characterisation, using bispectra as descriptors rather than learnable pooling primitives. Outside the equivariant-network setting, `PyBispectra` [Binns et al., 2025] computes bispectrum and bicoherence statistics of one-dimensional time-series signals for electrophysiology applications such as phase-amplitude coupling and time-delay estimation; the underlying 1D bispectrum is translation-invariant by construction, but the toolbox is not designed as a neural-network pooling layer.

Bispectra on homogeneous and non-homogeneous spaces. The selective G-bispectrum construction [Mataigne et al., 2024] reduces the cost of the G-bispectrum from $O(|G|^2)$ to $O(|G|)$ scalars on finite groups. Myers and Miolane [2025] extend the selective construction to the disk, where $\mathrm{SO}(2)$ acts on a non-homogeneous space. Kakarala [2012] treats the $\mathrm{SO}(3)$-on-$\mathrm{SO}(3)$ bispectrum and the closely related sphere case via Wigner-D irreps. The matrix (vector) bispectrum on S^2 is the homogeneous-space form on which our scalar contractions and CG power augmentation are built.

Invariant theory and scattering. The wavelet scattering transform [Mallat, 2012] provides translation-invariant representations via iterated modulus operations; Esteves et al. [2018] extend this to the sphere. Unlike the G-bispectrum, scattering representations are stable but generally incomplete—they discard inter-scale phase information. The G-bispectrum's completeness guarantee [Giannakis, 1989, Kakarala, 2012] is its distinguishing theoretical advantage.

4 The `bispectrum` Library

4.1 Design and implementation

Every G/domain pair is a `torch.nn.Module` named `{Group}on{Domain}` (e.g. `SO3onS2`) with a uniform API: `forward(f)` returns the selective invariants, `fourier(f)` the group Fourier coefficients, and `invert(beta)` the reconstructed signal up to group-action indeterminacy (where available). Modules default to $O(|G|)$ selective coefficients (`selective=False` restores the full $O(|G|^2)$ set), precompute CG matrices / DFT kernels / Bessel roots as non-learnable buffers, and depend only on PyTorch [Paszke et al., 2019], NumPy [Harris et al., 2020], and `torch_harmonics` [Bonev et al., 2023] (for `SO3onS2`). Test suite line coverage is 96.8%; a minimal usage example is in Appendix F.

4.2 Profiling

We profile the library on a single NVIDIA H100 80 GB GPU (PyTorch 2.10, batch = 16 for the forward pass and 4 for inversion, median wall-clock via `torch.utils.benchmark`); results are in Table 2. Coefficient-count scaling and additional plots are in Appendix B (Figure 4).

Table 3: PCam at matched $\sim$100K params, 10% training data, mean $\pm$ std over 3 seeds; all equivariant models share a C_8-equivariant DenseNet backbone.

Model	Params	Test AUC	Test ACC
Standard (augmented)	102K	$.896 \pm .010$	$.826 \pm .012$
Gated	136K	$.941 \pm .009$	$.870 \pm .004$
Selective Bispectrum (`CnonCn`)	128K	$.941 \pm .004$	$\mathbf{.872 \pm .006}$
Fourier-ELU	110K	$\mathbf{.945 \pm .004}$	$.855 \pm .006$
NormReLU	110K	$.942 \pm .004$	$.858 \pm .007$

The selective forward pass is sub-millisecond for finite abelian modules (C_n: 0.15 ms, $\mathbb{T}^2$: 0.08 ms), under 2 ms for $\mathrm{SO}(2)$/disk and under 1 ms for the octahedral group, making it negligible relative to backbone computation. $\mathrm{SO}(3)$ on S^2 at $\ell_{\max}$=16 takes 26.5 ms, dominated by the spherical harmonic transform (`torch_harmonics` [Bonev et al., 2023]). GPU throughput scales linearly with batch size, exceeding 10^7 samples/s for C_n and $\mathbb{T}^2$ (Appendix B). Six backends are tabulated; the seventh module, `SO2onS1`, subclasses `CnonCn` with $n = \ell_{\max}+1$ and has identical runtime to the corresponding `CnonCn` row at matched $|G|$.

5 Experiments

We validate the library on three tasks spanning discrete and continuous groups: 2D histopathology classification under C_8 symmetry (PCam, Section 5.1), 3D organ classification under octahedral symmetry (OrganMNIST3D, Section 5.2), and spherical digit classification under full $\mathrm{SO}(3)$ (Spherical MNIST, Section 5.3).

5.1 2D histopathology classification under cyclic symmetry

We evaluate five invariant pooling strategies on PatchCamelyon [Veeling et al., 2018] (PCam), 327K histopathology patches with binary metastasis labels. All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone following Veeling et al. [2018]; they differ only in the terminal invariant map: norm pooling, gated pooling, Fourier-ELU, and selective bispectrum via `CnonCn`(n=8). A non-equivariant standard CNN trained with data augmentation drawn uniformly from the symmetry group C_8 ("*augmented CNN*") serves as the non-equivariant baseline. We sweep growth rates to vary capacity (30K–1.6M params) and train on 10% of the PCam training set (an image-level subsample, not a group-orbit subsample, drawn before any group augmentation) with AdamW [Loshchilov and Hutter, 2019]. Full model descriptions are in Appendix C.

Table 3 reports multi-seed statistics at matched $\sim$100K params; the full Pareto sweep (AUC vs. parameter count, ranking, data efficiency) is Figure 7 in Appendix C.

At matched capacity ($\sim$100K params, 10% data), all four equivariant pooling methods substantially outperform the standard augmented CNN (AUC 0.896). Fourier-ELU leads slightly (0.945), followed by NormReLU (0.942), bispectrum and gated (both 0.941). At 1% data, the bispectrum leads all methods (0.912), outperforming NormReLU (0.873) and Fourier-ELU (0.876). This data-efficiency advantage narrows as the training set grows, consistent with complete invariants providing a stronger inductive bias when data is scarce.

5.2 3D organ classification under octahedral symmetry

To evaluate the `OctaonOcta` module for the chiral octahedral group O ($|O| = 24$), we design a controlled 3D experiment on OrganMNIST3D [Yang et al., 2023] (1,742 abdominal CT volumes, 28^3 voxels, 11 classes). All equivariant variants share an O-equivariant 3D ResNet backbone with exact voxel-permutation kernels. The four models compared are: a translation-equivariant standard 3D CNN trained with octahedral data augmentation (the "*augmented CNN*" baseline, 16K params); the same O-equivariant ResNet with terminal norm pool (375K params); with terminal max pool (374K params); and with terminal selective bispectrum via `OctaonOcta` (463K, 172 invariant scalars). Models are trained with AdamW [Loshchilov and Hutter, 2019] for 100 epochs with three seeds; rotation robustness is measured over all 24 octahedral rotations of the test set. Published baselines and wider-channel experiments are in Appendix D.

Table 4: OrganMNIST3D results (mean $\pm$ std over 3 seeds). Rot ACC and σ_{rot}: mean and std of accuracy across 24 octahedral test-set rotations.

Model	Params	Test ACC	Test AUC	Rot ACC	σ_{rot}
Standard 3D CNN	16K	$.601 \pm .017$	$.951 \pm .002$	.598	.0115
O-Equiv + Norm Pool	375K	$.568 \pm .099$	$.940 \pm .021$	.568	.0000
O-Equiv + Max Pool	374K	$.730 \pm .033$	$.972 \pm .005$	.730	.0000
O-Equiv + `OctaonOcta` Bispectrum	463K	$.726 \pm .027$	$.972 \pm .004$	.726	.0005†

†Nonzero σ_{rot} due to floating-point precision in CG contractions, not a theoretical invariance failure.

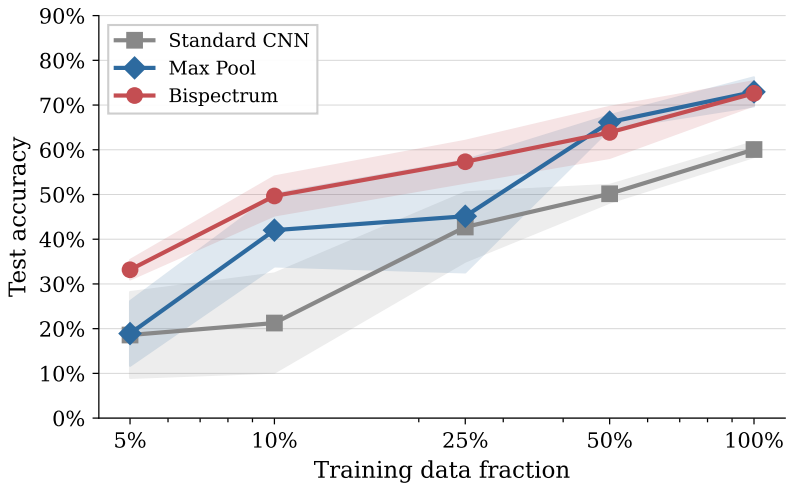

Figure 1: Test accuracy vs. training data fraction on OrganMNIST3D (`OctaonOcta` bispectrum, mean over 3 seeds; shaded bands: ± 1 std). The bispectrum provides a clear advantage in the low-data regime.

Table 4 and Figure 1 report the key findings.

All equivariant models achieve $\sigma_{\text{rot}} \leq 0.0005$ while the augmented CNN drops $\sim$2.5 pp under rotation. The bispectrum (72.6%) matches max pool (73.0%) and substantially outperforms norm pool (56.8%) and Regular SE(3) convolutions [Kuipers and Bekkers, 2023] (69.8%). At reduced data (Figure 1), the bispectrum's advantage widens: at 5% data it reaches 33.2% vs. 18.9% for max pool. At higher capacity (6M params, Appendix D), max pool overtakes the bispectrum, suggesting the completeness advantage is specific to the data-limited regime.

5.3 Spherical MNIST: SO(3) invariance on the sphere

The preceding experiments validate bispectral pooling for finite groups (C_8 and the octahedral group O). To evaluate the library's continuous group support, we test `SO3onS2`—the augmented selective $\mathrm{SO}(3)$ invariant on the 2-sphere (Section 6)—on Spherical MNIST [Cohen et al., 2018], the standard benchmark for rotation invariance on S^2.

Setup. MNIST digits are stereographically projected onto a 64×128 equiangular grid; we compute Φ_{sel} at $\ell_{\max} = 15$ ($N = 384$ live real components, stored as 384 complex tensors by the library), apply sign-preserving $\mathrm{sign}(x)\log(1+|x|)$ to real and imaginary parts ($2N = 768$ real channels; one channel per bispectral entry is structurally zero by parity but we feed both since the log transform is per-channel injective and cheap), and feed a 3-layer MLP (232K params). Controls: a power-spectrum MLP at 11K params (equally invariant but *incomplete*, discarding inter-harmonic phase), a matched 166K-param version isolating capacity vs. completeness, and a non-invariant CNN directly on the equirectangular image at 185K params. Following Cohen et al. [2018], we evaluate under three protocols NR/NR, R/R, and the most challenging NR/R; each run uses three seeds, and rotation robustness σ_{rot} is measured over 10 random $\mathrm{SO}(3)$ rotations of the test set.

Table 5 and Figure 2 report the results.

Table 5: Spherical MNIST results (mean $\pm$ std over 3 seeds). Reference numbers are from Cohen et al. [2018]. σ_{rot}: standard deviation of accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models).

Model	Params	NR/NR ↑	R/R ↑	NR/R ↑	σ_{rot} ↓
Cohen planar CNN	—	0.98	0.23	0.11	—
Cohen spherical CNN	—	0.96	0.95	0.94	—
Standard CNN	185K	0.991±0.000	0.781±0.003	0.138±0.003	0.044
Power Spec. + MLP	11K	0.780±0.002	0.780±0.002	0.777±0.001	3×10^{-4}
Power Spec. + MLP (matched)	166K	0.794±0.011	0.794±0.011	0.792±0.010	3×10^{-4}
`SO3onS2` **Bispectrum + MLP (ours)**	232K	**0.950±0.000**	**0.949±0.002**	**0.951±0.001**	$\mathbf{1\times10^{-4}}$

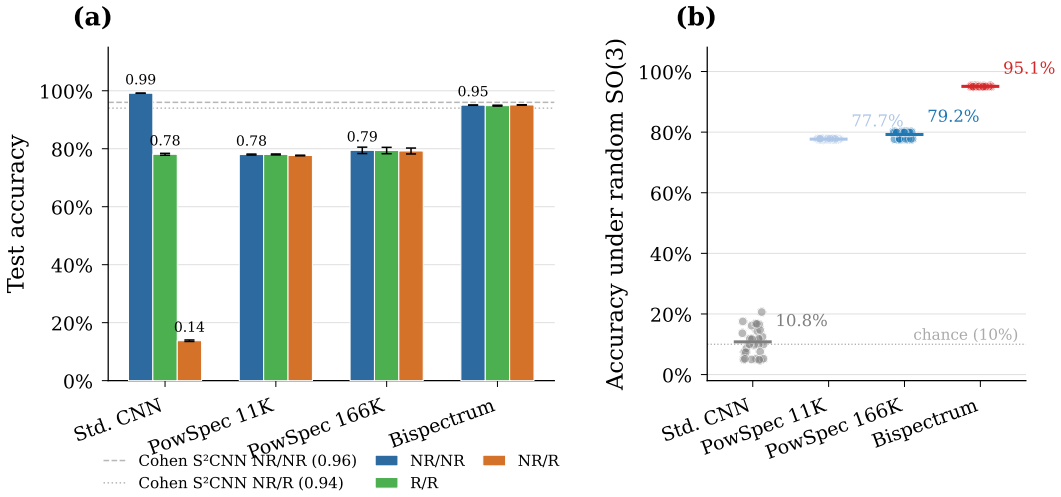

Figure 2: Spherical MNIST analysis (`SO3onS2` selective bispectrum). **(a)** Cohen et al. protocol: NR/NR, R/R, and NR/R accuracy. Dashed lines show Cohen's spherical CNN reference. **(b)** Accuracy under 10 random $\mathrm{SO}(3)$ rotations (NR-trained models, 30 points per model = 10 rotations $\times$ 3 seeds). Scaling the power spectrum MLP from 11K to 166K parameters yields only +1.4pp; the `SO3onS2` bispectrum at 232K reaches 95.0%.

The bispectrum achieves near-identical accuracy across all three protocols ($\sigma_{\text{rot}}\sim10^{-4}$), confirming true $\mathrm{SO}(3)$ invariance; the standard CNN collapses from 99.1% (NR/NR) to 13.8% (NR/R), showing that augmentation alone does not confer invariance. The power spectrum is equally invariant but reaches only 78%—a 17pp deficit that persists after scaling the MLP to 166K params (+1.4pp), consistent with the power spectrum's incompleteness (rather than MLP capacity) being the bottleneck. Cohen's spherical CNN [Cohen et al., 2018] reaches 96%/95%/94% with learned equivariant convolutions; our single-pass bispectrum (95.0%/94.9%/95.1%) is 1–2pp lower but has a tighter invariance gap (0.1pp vs. 2.0pp).

6 Augmented selective $\mathrm{SO}(3)$ invariant on S^2

6.1 Setup and construction

A real signal $f : S^2 \to \mathbb{R}$ band-limited at degree L has spherical harmonic coefficients $\mathbf{F}_\ell \in \mathbb{C}^{2\ell+1}$, $\ell = 0, \ldots, L$, transforming as $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$ under $g \in \mathrm{SO}(3)$; the reality condition leaves $(L+1)^2$ real degrees of freedom (Appendix A.1). The scalar bispectral coefficient at a CG-admissible triple (ℓ_1, ℓ_2, ℓ) is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,m_2,m \\ m_1+m_2=m}} C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}\, a^{m_1}_{\ell_1}\, a^{m_2}_{\ell_2}\, \overline{a^{m}_{\ell}}, \tag{3}$$

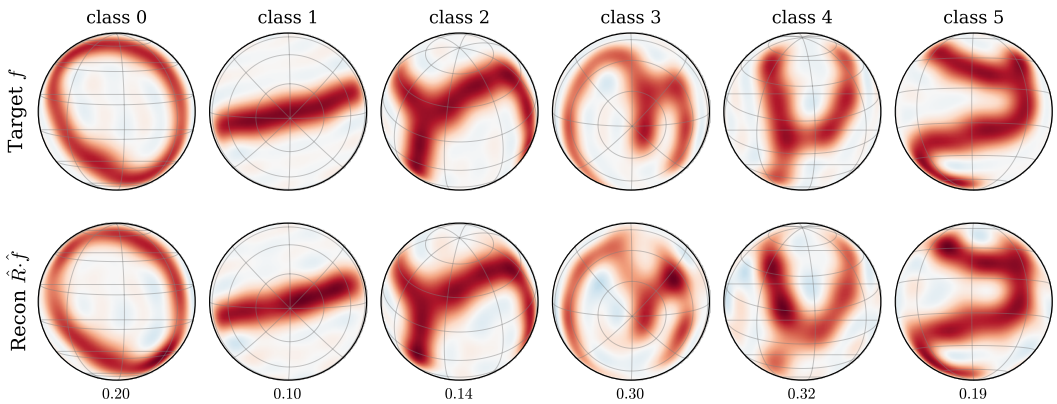

Figure 3: Φ_{sel} reconstruction (proof of concept). Top: target f (digit classes 0–5). Bottom: $\hat{R}\cdot\hat{f}$, recovered from $\Phi_{\text{sel}}(f)$ by gradient descent and Procrustes-aligned. Numbers: relative image residual $\|\hat{R}\cdot\hat{f}-f\|/\|f\|$, median $\approx$0.20 across the six classes (raw pre-alignment residual $\approx$1.3).

the specialisation of (2) to $\mathrm{SO}(3)/S^2$. The full set has $\Theta(L^3)$ scalars; the orbit space $\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)$ has dimension $(L{+}1)^2-3$, so any smooth complete invariant requires $\Omega(L^2)$ components (Proposition 7), which the Mataigne chain $\beta_{1,\ell,\ell+1}$ ($\sim 3L$ scalars) cannot achieve.

We define $\Phi_{\text{sel}}(f)$ as the output of the augmented selective index construction of Algorithm 2 (Appendix A.3). For each degree ℓ it collects (i) a size-$(2\ell{+}1)$ linear bootstrap block $\mathcal{T}_\ell$ of scalar bispectrum entries, (ii) mandatory even self-couplings $\beta_{\ell,\ell,\ell'}$, and (iii) a greedily chosen set of CG-power scalars $P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1}\otimes\mathbf{F}_{\ell_2})|_\ell\|^2$ (the bootstrap block alone is rank-deficient on real signals, see Remark 18). Because Φ_{sel} mixes bispectrum entries with CG-power scalars (a degree-4 invariant), we call it an *augmented selective invariant* rather than a bispectrum; the phrase "selective bispectrum" refers strictly to the bispectral component $\mathcal{S}_\beta$. For degrees $\ell \le L_{\text{seed}} := \min(L,4)$ we additionally include the full seed $\mathcal{S}_{\text{seed}}$. The total size is $|\Phi_{\text{sel}}| = \Theta(L^2)$ (Proposition 11); Algorithm 1 summarises the bispectral-triple selection, and the full index set—including self-couplings and CG power—is laid out in Algorithm 2. Concretely, at $L=15$ (resp. $L=16$) the library outputs $|\Phi_{\text{sel}}| = 384$ (resp. 430) real scalars; the full bispectrum at $L=15$ has 2,056 admissible triples (cf. Table 7).

Algorithm 1 Selected bispectral triples $\mathcal{S}_\beta$ for the augmented $\mathrm{SO}(3)$ invariant (BFS form; mirrors Mataigne et al. [2024], Alg. 1). The full index set — bispectral triples, mandatory self-couplings, and CG-power scalars — is Algorithm 2 in Appendix A.3.

```
Input: band-limit L, seed cutoff L_seed = min(L, 4)
L_β ← all admissible triples (ℓ1, ℓ2, ℓ) with ℓ ≤ L_seed {seed block S_seed}
for ℓ = L_seed+1, ..., L do
    append the bootstrap block T_ℓ to L_β {2ℓ+1 entries linear in F_ℓ; tabulated in Appendix A.3}
end for
return {β_{ℓ1,ℓ2,ℓ}(f) : (ℓ1, ℓ2, ℓ) ∈ L_β}
```

6.2 Empirical proof of concept: bispectrum reconstruction

To test whether Φ_{sel} retains enough information to recover signals up to rotation, we run a direct inversion experiment on Spherical MNIST digits band-limited at $L=12$.[1] For each source signal $f: S^2\to\mathbb{R}$ we apply a uniform random rotation $R\in\mathrm{SO}(3)$ to form the target $R\cdot f$, compute $\Phi_{\text{sel}}(R\cdot f)$, and recover $\hat{f}$ by gradient descent from a random Gaussian initialisation, minimising $\|\Phi_{\text{sel}}(\hat{f})-\Phi_{\text{sel}}(R\cdot f)\|^2/\|\Phi_{\text{sel}}(R\cdot f)\|^2$ (Adam, multi-restart, cosine-annealed LR). The orbit ambiguity is then resolved by Procrustes-style optimisation on a quaternion parameterisation: we fit $\hat{R}\in\mathrm{SO}(3)$ minimising $\|\hat{R}\cdot\hat{f}-R\cdot f\|^2$.

[1] We use $L{=}12$ here rather than the classifier's $L{=}15$ to keep the reconstruction residual safely below the SHT discretisation floor at the $64{\times}128$ grid; protocol details are in Appendix E.5.

Takeaways. Bispectrum residuals reach $\|\Delta\Phi\|/\|\Phi\| \sim 10^{-4}$–$10^{-3}$ (below the SHT-discretisation invariance noise floor of $\sim 8\times 10^{-3}$ at this grid); the aligned image residuals $\|\hat{R}\cdot\hat{f} - f\|/\|f\| \sim 0.1$–$0.3$ are consistent with local-optimisation error or conditioning effects, and we did not observe a systematic missing-information failure mode, but the experiment does not rule out incompleteness or near-collisions in feature space. Visually, $\hat{R}\cdot\hat{f}$ aligns with f up to rotation across all six digit classes shown (Figure 3; the same behaviour holds across our 8-digit sweep, see Appendix E.5). We read this as proof-of-concept evidence that Φ_{sel} preserves substantial orbit information on band-limited signals; a formal completeness proof remains open.

7 Discussion and Conclusion

We have presented `bispectrum`, a PyTorch library that makes selective G-bispectral invariants practical, covering seven group/domain pairs through a uniform `nn.Module` interface, and we contribute a $\Theta(L^2)$ selective $\mathrm{SO}(3)$ invariant on S^2 (Section 6). The selective forward pass adds negligible overhead ($<$1 ms for finite-group modules) and delivers numerically exact rotation invariance with a consistent data-efficiency advantage over incomplete alternatives in the low-data / moderate-capacity regime.

When not to use the bispectrum. At high model capacity (Sec. 5), incomplete alternatives like max pooling can match or surpass bispectral pooling, and each module is invariant to a *specific* group only—data augmentation or flexible-symmetry equivariant networks fit better when the relevant symmetry is multi-group or unknown.

Limitations and future work. Inversion from Φ_{sel} is not shipped as a library API, and a formal completeness proof, a characterisation of the degenerate set, and numerical-conditioning analysis at large band-limits remain open. We do not compare against tensor-product architectures (e3nn, MACE) or wavelet scattering. Natural next steps are $\mathrm{SO}(3)$ inversion and cryo-EM / molecular applications.

References

Ilyes Batatia, Mario Geiger, Jose Munoz, Tess Smidt, Lior Silberman, and Christoph Ortner. A general framework for equivariant neural networks on reductive Lie groups, 2023. URL `https://arxiv.org/abs/2306.00091`.

Tamir Bendory, Dan Edidin, Josh Katz, and Shay Kreymer. Orbit recovery for spherical functions. *arXiv preprint arXiv:2508.02674*, 2025. URL `https://arxiv.org/abs/2508.02674`.

Thomas S. Binns, Franziska Pellegrini, Tin Jurhar, Tien D. Nguyen, Richard M. Köhler, and Stefan Haufe. PyBispectra: A toolbox for advanced electrophysiological signal processing using the bispectrum. *Journal of Open Source Software*, 10(112):8504, 2025. doi: 10.21105/joss.08504. URL `https://doi.org/10.21105/joss.08504`.

Boris Bonev, Thorsten Kurth, Christian Hundt, Jaideep Pathak, Maximilian Balu, Karthik Kashinath, and Animashree Anandkumar. Spherical Fourier neural operators: Learning stable dynamics on the sphere. In *International Conference on Machine Learning (ICML)*, 2023.

Gabriele Cesa, Leon Lang, and Maurice Weiler. A program to build E(N)-equivariant steerable CNNs. In *International Conference on Learning Representations (ICLR)*, 2022. URL `https://openreview.net/forum?id=WE4qe9xlnQw`.

Arthur Chevalley, Valentin Oreiller, Julien Fageot, John O. Prior, Vincent Andrearczyk, and Adrien Depeursinge. A bispectral 3D U-Net for rotation robustness in medical segmentation. In *Topology- and Graph-Informed Imaging Informatics*, pages 43–54. Springer Nature Switzerland, 2024. doi: 10.1007/978-3-031-73967-5_5. URL `https://doi.org/10.1007/978-3-031-73967-5_5`.

Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning (ICML)*, pages 2990–2999, 2016.

Taco S. Cohen, Mario Geiger, Jonas Köhler, and Max Welling. Spherical CNNs. In *International Conference on Learning Representations (ICLR)*, 2018.

Persi Diaconis and Daniel Rockmore. Efficient computation of the Fourier transform on finite groups. *Journal of the American Mathematical Society*, 3(2):297–332, April 1990. doi: 10.1090/S0894-0347-1990-1030655-4. URL https://www.ams.org/journals/jams/1990-03-02/S0894-0347-1990-1030655-4/.

Dan Edidin and Matthew Satriano. Orbit recovery for band-limited functions. *SIAM Journal on Applied Algebra and Geometry*, 8(3):733–755, 2024. doi: 10.1137/23M1577808. URL https://arxiv.org/abs/2306.00155.

Babak Ehteshami Bejnordi, Mitko Veta, Paul Johannes van Diest, Bram van Ginneken, Nico Karssemeijer, Geert Litjens, Jeroen A. W. M. van der Laak, and the CAMELYON16 Consortium. Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA*, 318(22):2199–2210, 2017. doi: 10.1001/jama.2017.14585.

Carlos Esteves, Christine Allen-Blanchette, Ameesh Makadia, and Kostas Daniilidis. Learning SO(3) equivariant representations with spherical CNNs. In *European Conference on Computer Vision (ECCV)*, pages 52–68. Springer, 2018.

Mario Geiger and Tess Smidt. e3nn: Euclidean neural networks. *arXiv preprint arXiv:2207.09453*, 2022. URL https://arxiv.org/abs/2207.09453.

Georgios B. Giannakis. Signal reconstruction from multiple correlations: frequency- and time-domain approaches. *Journal of the Optical Society of America A*, 6(5):682–697, May 1989. doi: 10.1364/JOSAA.6.000682. URL https://opg.optica.org/josaa/abstract.cfm?URI=josaa-6-5-682.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.

Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

Ramakrishna Kakarala. *Triple Correlation on Groups*. PhD thesis, University of California, Irvine, 1992.

Ramakrishna Kakarala. The bispectrum as a source of phase-sensitive invariants for Fourier descriptors: A group-theoretic approach. *Journal of Mathematical Imaging and Vision*, 44(3):341–353, 2012. doi: 10.1007/s10851-012-0330-6. URL https://arxiv.org/abs/0902.0196.

Thijs P. Kuipers and Erik J. Bekkers. Regular SE(3) group convolutions for volumetric medical image analysis. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, pages 252–261. Springer Nature Switzerland, 2023. doi: 10.1007/978-3-031-43898-1_25. URL https://doi.org/10.1007/978-3-031-43898-1_25.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Stéphane Mallat. Group invariant scattering. *Communications on Pure and Applied Mathematics*, 65(10):1331–1398, 2012. doi: 10.1002/cpa.21413.

Simon Mataigne, Johan Mathe, Sophia Sanborn, Christopher Hillar, and Nina Miolane. The selective G-bispectrum and its inversion: Applications to G-invariant networks. In *Advances in Neural Information Processing Systems*, volume 37, pages 115682–115711. Neural Information Processing Systems Foundation, Inc., 2024. doi: 10.52202/079017-3674. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/d1a1e8713fcd5626656553c82f7c3b26-Abstract-Conference.html.

Sophia Myers and Nina Miolane. The selective disk bispectrum and its inversion, with application to multi-reference alignment. *arXiv preprint arXiv:2511.19706*, 2025. URL https://arxiv.org/abs/2511.19706.

Valentin Oreiller, Julien Fageot, Vincent Andrearczyk, John O. Prior, and Adrien Depeursinge. Robust multi-organ nucleus segmentation using a locally rotation invariant bispectral U-Net. In *Proceedings of The 5th International Conference on Medical Imaging with Deep Learning*, volume 172 of *Proceedings of Machine Learning Research*, pages 929–943. PMLR, 2022. URL https://proceedings.mlr.press/v172/oreiller22a.html.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Sophia Sanborn, Christian Shewmake, Bruno Olshausen, and Christopher Hillar. Bispectral neural networks. In *International Conference on Learning Representations (ICLR)*, 2023.

Mateus Sangalli, Samy Blusseau, Santiago Velasco-Forero, and Jesús Angulo. Moving frame net: SE(3)-equivariant network for volumes. In *Proceedings of the 1st NeurIPS Workshop on Symmetry and Geometry in Neural Representations*, volume 197 of *Proceedings of Machine Learning Research*, pages 81–97. PMLR, 2023. URL https://proceedings.mlr.press/v197/sangalli23a.html.

Bastiaan S. Veeling, Jasper Linmans, Jim Winkens, Taco Cohen, and Max Welling. Rotation equivariant CNNs for digital pathology. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 210–218. Springer, 2018.

Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

Dmitrii Zhemchuzhnikov and Sergei Grudinin. ILPO-NET: Network for the invariant recognition of arbitrary volumetric patterns in 3D. In *Machine Learning and Knowledge Discovery in Databases. Research Track*, pages 352–368. Springer Nature Switzerland, 2024. doi: 10.1007/978-3-031-70359-1_21. URL https://doi.org/10.1007/978-3-031-70359-1_21.

Dmitrii Zhemchuzhnikov and Sergei Grudinin. On the Fourier analysis in the SO(3) space: the EquiLoPO network. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://openreview.net/forum?id=LvTSvdiSwG.

A Selective $\mathrm{SO}(3)$ Bispectrum – Structural Background

This appendix gives the structural background for the construction of Section 6: the S^2 band-limited setup and notation (Section A.1); the orbit-space dimension lower bound $(L{+}1)^2 - 3$ that motivates the $\Theta(L^2)$ target (Section A.2); the explicit selective index set together with the implementation-grade BFS variant (Algorithm 2) and per-degree counting result (Section A.3); and the parity structure of the scalar bispectrum that dictates the exclusion of certain odd-parity triples in $\mathcal{T}_\ell$ (Section A.4). A short comparison with prior bispectral constructions (Section A.5) and a pixel-space complexity table (Section A.6) close the appendix. We do not claim a closed-form completeness proof in this revision; the empirical reconstruction of Figure 3 is the proof-of-concept evidence the construction is supported by.

Per-degree entry allocation: the full seed $\mathcal{S}_{\mathrm{seed}}$ handles $\ell \leq L_{\mathrm{seed}} = 4$ in constant size; for $\ell \geq 5$, Algorithm 2 allocates $2\ell{+}1$ bispectral rows plus $O(\ell)$ self-coupling and CG-power entries, giving $|\Phi_{\mathrm{sel}}| = \Theta(L^2)$ total (matching the orbit-space lower bound $(L{+}1)^2 - 3$ of Proposition 7; exact per-L counts in Table 7).

A.1 Setup and notation

Definition 1 (Band-limited signal on S^2)**.** A signal $f : S^2 \to \mathbb{R}$ is *band-limited* at degree L if its spherical harmonic expansion

$$f(\theta, \varphi) = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} a_\ell^m \, Y_\ell^m(\theta, \varphi) \tag{4}$$

has $a_\ell^m = 0$ for all $\ell > L$. We write $\mathbf{F}_\ell = (a_\ell^{-\ell}, \ldots, a_\ell^\ell) \in \mathbb{C}^{2\ell+1}$ for the coefficient vector at degree ℓ.

For real f, the conjugation symmetry $a_\ell^{-m} = (-1)^m \overline{a_\ell^m}$ holds, so $\mathbf{F}_\ell$ has $2\ell + 1$ real degrees of freedom. The total signal dimension is $\dim = \sum_{\ell=0}^{L}(2\ell + 1) = (L + 1)^2$.

Definition 2 (Rotation action)**.** Under $g \in \mathrm{SO}(3)$, the spherical harmonic coefficients transform as $\mathbf{F}_\ell \mapsto D^\ell(g)\, \mathbf{F}_\ell$, where D^ℓ is the $(2\ell + 1)$-dimensional Wigner D-matrix.

In the irrep notation of Mataigne et al. [2024], $\{\rho_\ell\}_{\ell \geq 0}$ are the (unitary) irreducible representations of $\mathrm{SO}(3)$ with V_{ρ_ℓ} the $(2\ell{+}1)$-dimensional space of degree-ℓ spherical harmonics and $\rho_\ell(g) = D^\ell(g)$; the Fourier coefficient of f at ρ_ℓ is $\mathcal{F}(f)_{\rho_\ell} = \mathbf{F}_\ell$. We use $\mathbf{F}_\ell$ as the working symbol throughout.

Definition 3 (Bispectrum on S^2)**.** For each triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$, the *bispectral coefficient* is

$$\beta_{\ell_1,\ell_2,\ell} = \sum_{\substack{m_1,\, m_2,\, m \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \, \overline{a_\ell^m}, \tag{5}$$

where $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$ denotes the Clebsch–Gordan coefficient $\langle \ell_1, m_1;\, \ell_2, m_2 \mid \ell, m \rangle$. Each $\beta_{\ell_1,\ell_2,\ell}$ is a scalar; for real f it is real when $\ell_1{+}\ell_2{+}\ell$ is even and purely imaginary when $\ell_1{+}\ell_2{+}\ell$ is odd (Lemma 12).

Definition 3 is the specialization of the general G-bispectrum (eq. (2)) to the case $G = \mathrm{SO}(3)$ acting on S^2, where the irreps are Wigner-D matrices and the Fourier coefficients are spherical harmonic coefficients a_ℓ^m.

Definition 4 (CG power spectrum, optional diagnostic)**.** For a triple (ℓ_1, ℓ_2, ℓ) satisfying the triangle inequality, write $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})\big|_\ell$ for the projection of the Kronecker product $\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2}$ onto the D^ℓ-isotypic component, realised by contracting with the CG coefficients $C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2}$. The *CG power spectrum entry* is the squared norm of this projection,

$$P_{\ell_1,\ell_2,\ell} := \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2 = \sum_{m=-\ell}^{\ell} \Bigg| \sum_{\substack{m_1, m_2 \\ m_1+m_2=m}} C^{\ell,\, m}_{\ell_1,\, m_1;\, \ell_2,\, m_2} \; a_{\ell_1}^{m_1} \, a_{\ell_2}^{m_2} \Bigg|^2. \tag{6}$$

This is a degree-4, real-valued, $\mathrm{SO}(3)$-invariant polynomial in the signal coefficients. The library appends a $O(L^2)$ family of CG power scalars to the selective bispectrum unconditionally for numerical robustness on the real-signal slice (Remark 18); they leave the asymptotic output size unchanged.

Proposition 5 (Size of the full bispectrum)**.** *The number of admissible triples* (ℓ_1, ℓ_2, ℓ) *with* $0 \leq \ell_1, \ell_2, \ell \leq L$ *satisfying the triangle inequality* $|\ell_1 - \ell_2| \leq \ell \leq \ell_1 + \ell_2$ *is* $\Theta(L^3)$.

Proof. For fixed ℓ_1 and ℓ_2, the triangle inequality admits $\min(\ell_1 + \ell_2, L) - |\ell_1 - \ell_2| + 1$ valid values of ℓ. Summing over $(\ell_1, \ell_2) \in [0, L]^2$ gives $N_{\text{full}} = \Theta(L^3)$. □

Remark 6 (Invariance). Each $\beta_{\ell_1,\ell_2,\ell}$ is $\mathrm{SO}(3)$-invariant: the Wigner matrices satisfy $D^{\ell_1} \otimes D^{\ell_2} = \bigoplus_\ell D^\ell$ under the CG decomposition, so the triple product contracted with CG coefficients is unchanged by $\mathbf{F}_\ell \mapsto D^\ell(g)\mathbf{F}_\ell$.

A.2 Why $\mathcal{O}(L)$ is impossible

Proposition 7 (Dimensional lower bound)**.** *Any* smooth or polynomial *complete* $\mathrm{SO}(3)$*-invariant for band-limited signals on* S^2 *at band-limit* L *must have codomain of dimension at least* $(L{+}1)^2{-}3$.

Proof. Let $\Phi : \mathbb{R}^{(L+1)^2} \to \mathbb{R}^k$ be a smooth (or polynomial) $\mathrm{SO}(3)$-invariant map that separates orbits on a Zariski-open dense subset $U \subset \mathbb{R}^{(L+1)^2}$. Smoothness implies Φ has a well-defined generic rank $r := \max_{x\in U} \operatorname{rank} d\Phi_x$; on the open dense subset where this rank is attained, Φ is locally a submersion onto an r-dimensional submanifold, and orbit-separation forces $r \geq \dim(\mathbb{R}^{(L+1)^2}/\mathrm{SO}(3)) = (L+1)^2 - \dim \mathrm{SO}(3) = (L+1)^2 - 3$ (generic-rank theorem for $\mathrm{SO}(3)$ acting on V_L; cf. Tarski–Seidenberg in the polynomial case). Hence $k \geq r \geq (L{+}1)^2 - 3 = \Omega(L^2)$. □

Remark 8. The chain $\beta_{1,\ell,\ell+1}$ for $\ell = 0, \ldots, L{-}1$ provides only $\sim 3L$ scalar constraints (since each chain entry couples $\mathbf{F}_1 \in \mathbb{C}^3$ with one known $\mathbf{F}_\ell$). This cannot recover $(L+1)^2 \approx L^2$ unknowns.

Remark 9 (Smoothness assumption). The smoothness/polynomial qualifier is essential: pathological non-measurable invariants (e.g. a fixed orbit-section indexed by the axiom of choice) can in principle have codomain dimension 1 but are not implementable. All practical bispectral or spectral invariants are polynomial, so the proposition applies.

A.3 The selective index set

For each target degree ℓ, the library's selective invariant allocates $2\ell{+}1$ linear bispectral entries, plus quadratic/cubic self-couplings and CG-power scalars for numerical robustness on the real-signal slice. Combined with a constant-size full-bispectrum seed at degrees $\leq L_{\mathrm{seed}} = \min(L, 4)$, this gives an $O(L^2)$-sized invariant matching the orbit-space lower bound of Proposition 7.

Definition 10 (Augmented selective $\mathrm{SO}(3)$-on-S^2 invariant)**.** Set $L_{\mathrm{seed}} := \min(L, 4)$. Let $\mathcal{S}_{\mathrm{seed}}$ denote the bispectral triples at degrees $\ell \leq L_{\mathrm{seed}}$ returned by Algorithm 2's seed phase: the pre-prune set collects all admissible triples $\{(\ell_1, \ell_2, \ell) : 0 \leq \ell_1 \leq \ell_2 \leq L_{\mathrm{seed}}, |\ell_1{-}\ell_2| \leq \ell \leq \min(\ell_1{+}\ell_2, L_{\mathrm{seed}})\}$ (42 triples at $L_{\mathrm{seed}} = 4$); Algorithm 2 then parity-prunes entries that vanish identically on real signals (Lemma 12), deduplicates, and caps each degree to its per-degree budget. The surviving bispectral count at $L = 4$ is 24 (see Table 7, column "Bispectral"), and $|\mathcal{S}_{\mathrm{seed}}|$ is constant in L for $L \geq 4$. For each intermediate target $4 \leq \ell \leq 7$ Algorithm 2 uses the small explicit linear block $\mathcal{T}_\ell^{\mathrm{small}}$ (Table 6); for each high-degree target $\ell \geq 8$ the *bootstrap block* $\mathcal{T}_\ell$ is

$$\begin{aligned}\mathcal{T}_\ell := &\{(a, \ell, \ell - a) : 1 \leq a \leq \ell{-}1\} \\ &\cup \big\{(a, \ell, \ell - a + 1) : 2 \leq a \leq \ell{-}1,\ a \neq (\ell+1)/2\big\} \\ &\cup \{(a, \ell - a, \ell) : 1 \leq a \leq 4\} \ \cup\ \mathcal{Z}_\ell,\end{aligned} \tag{7}$$

where $\mathcal{Z}_\ell = \emptyset$ for even ℓ and $\mathcal{Z}_\ell = \{(2, \ell{-}1, \ell)\}$ for odd $\ell \geq 9$. The exclusion $a \neq (\ell+1)/2$ for odd ℓ removes the self-conjugate triple $(r, 2r-1, r)$ at $\ell = 2r-1$, which has a repeated index and vanishes identically on real signals (Proposition 14); the compensating row $(2, \ell{-}1, \ell) \in \mathcal{Z}_\ell$ has all-distinct indices and does not overlap the chain family (which only covers pairs $(a, \ell{-}a)$ with $a \leq 4$). Size bookkeeping: $(\ell{-}1) + (\ell{-}2) + 4 = 2\ell{+}1$ for even ℓ, and $(\ell{-}1) + (\ell{-}3) + 4 + 1 = 2\ell{+}1$ for odd $\ell \geq 9$; the first two families differ in the last index ($\ell{-}a$ vs. $\ell{-}a+1$), and $\mathcal{Z}_\ell$ at $a = 2$ has second coordinate $\ell{-}1$ rather than $\ell{-}2$, so all three subfamilies are disjoint.

The *augmented selective* $\mathrm{SO}(3)$*-on-*S^2 *invariant* Φ_{sel} is the output of Algorithm 2: bispectral triples $\mathcal{S}_\beta = \mathcal{S}_{\mathrm{seed}} \cup \bigcup_{\ell=5}^{L} \mathcal{T}_\ell$ (with $\mathcal{T}_\ell = \mathcal{T}_\ell^{\mathrm{small}}$ for $5 \leq \ell \leq 7$), mandatory even self-couplings $\beta_{\ell,\ell,\ell'}$ (ℓ' even, $2 \leq \ell' \leq \ell$), and CG-power scalars $P_{\ell_1,\ell_2,\ell} = \|(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell\|^2$ (Definition 4) chosen greedily for per-degree Jacobian rank. Mathematically, $\Phi_{\mathrm{sel}} : V_L \to \mathbb{R}^N$ has N live real scalar components: by Lemma 12, each bispectral entry contributes one real component on the real-signal slice (real for even parity, purely imaginary for odd), and CG-power entries are real-valued degree-4 invariants. The PyTorch implementation stores each scalar in a complex tensor, so downstream layers receive $2N$ real channels after splitting real and imaginary parts; for bispectral entries exactly one of the two channels is structurally zero by parity, and for CG-power entries the imaginary channel is structurally zero.

Table 6: Explicit low-degree bootstrap blocks $\mathcal{T}_\ell^{\text{small}}$ for $4 \le \ell \le 7$, copied from `bispectrum.so3_on_s2._small_linear_bootstrap_block`. Each triple (ℓ_1, ℓ_2, ℓ_3) denotes $\beta_{\ell_1,\ell_2,\ell_3}$; all are admissible and have ℓ in at least one slot. Entry counts $|\mathcal{T}_4^{\text{small}}| = 10$, $|\mathcal{T}_5^{\text{small}}| = 11$, $|\mathcal{T}_6^{\text{small}}| = 13$, $|\mathcal{T}_7^{\text{small}}| = 15$ match the per-degree budgets. $\mathcal{T}_4^{\text{small}}$ is implementation-only: Algorithm 2 uses it as a ranked candidate list at $\ell = 4$, deduplicated against the seed $\mathcal{S}_{\text{seed}}$ and capped to a budget of 10 entries; $\mathcal{T}_5^{\text{small}}, \mathcal{T}_6^{\text{small}}, \mathcal{T}_7^{\text{small}}$ are the formal bootstrap blocks at $\ell \in \{5, 6, 7\}$.

ℓ	$\mathcal{T}_\ell^{\text{small}}$ (each entry a triple (ℓ_1, ℓ_2, ℓ_3))
4	$(1,3,4)$, $(2,2,4)$, $(2,3,4)$, $(3,3,4)$, $(1,4,3)$, $(2,4,2)$, $(3,4,1)$, $(2,4,3)$, $(3,4,2)$, $(3,4,3)$
5	$(1,4,5)$, $(2,3,5)$, $(2,4,5)$, $(3,4,5)$, $(1,5,4)$, $(2,5,3)$, $(3,5,2)$, $(4,5,1)$, $(2,5,4)$, $(3,5,4)$, $(4,5,4)$
6	$(1,5,6)$, $(2,4,6)$, $(3,3,6)$, $(3,4,6)$, $(1,6,5)$, $(2,6,4)$, $(3,6,3)$, $(4,6,2)$, $(5,6,1)$, $(2,6,5)$, $(3,6,5)$, $(4,6,5)$, $(5,6,5)$
7	$(1,6,7)$, $(2,5,7)$, $(3,4,7)$, $(4,5,7)$, $(1,7,6)$, $(2,7,5)$, $(3,7,4)$, $(4,7,3)$, $(5,7,2)$, $(6,7,1)$, $(2,7,6)$, $(3,7,6)$, $(4,7,6)$, $(5,7,6)$, $(6,7,6)$

Table 7: Coefficient counts of Φ_{sel} at representative band-limits. "Bispectral" counts the triples in $\mathcal{S}_\beta$ (including mandatory even self-couplings); "CG-power" counts entries in $\mathcal{S}_P$; "Total N" is $|\Phi_{\text{sel}}|$ (live real components) and matches the library's `output_size`. "Stored channels" counts the complex-tensor storage ($2N$) seen by the SMNIST classifier; one channel per bispectral entry is structurally zero by parity, and the imaginary channel of each CG-power entry is structurally zero.

L	Bispectral	CG-power	Total $N = \lvert\Phi_{\text{sel}}\rvert$	Stored channels ($2N$)
4	24	10	34	68
5	37	17	54	108
15	307	77	384	768
16	348	82	430	860

Proposition 11 (Output size)**.** *For every $L \ge 4$, Algorithm 2 produces a $\Theta(L^2)$-sized invariant: the per-degree bootstrap block contributes at most $2\ell+1$ bispectral triples (Eq.* (7) *for $\ell \ge 8$ plus Table 6 for $5 \le \ell \le 7$), and the self-coupling and CG-power augmentations add $O(\ell)$ per degree, giving a total of $\Theta(L^2)$ real scalars. Per-L counts are listed in Table 7; both panels match the orbit-space lower bound $(L+1)^2 - 3$ of Proposition 7 up to a constant factor.*

A.4 Parity structure of the bispectrum

The selective construction of Section 6 exploits two structural facts about the scalar bispectrum $\beta_{\ell_1,\ell_2,\ell}$: a parity rule that fixes whether each entry is real or purely imaginary on real signals (Lemma 12) and the resulting identical vanishing of certain odd-parity entries with a repeated index (Proposition 14). The latter dictates the explicit exclusion in $\mathcal{T}_\ell$ at odd target degrees, and the former is the source of the rank deficit that motivates the CG-power augmentation of Algorithm 2 (Remark 18).

Lemma 12 (Parity of the bispectrum)**.** *For any signal (real or complex):*

$$\beta_{\ell_1,\ell_2,\ell}(T_R \cdot f) = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{8}$$

For real *signals, there is an independent constraint:*

$$\overline{\beta_{\ell_1,\ell_2,\ell}(f)} = (-1)^{\ell_1+\ell_2+\ell}\, \beta_{\ell_1,\ell_2,\ell}(f). \tag{9}$$

Together:

(a) If $\ell_1 + \ell_2 + \ell$ is even: β is real-valued (by (9)*) and T_R-invariant (by* (8)*).*
(b) If $\ell_1 + \ell_2 + \ell$ is odd: β is purely imaginary (by (9)*) and $\beta(T_R \cdot f) = -\beta(f)$, so $\operatorname{Im}\beta$ flips sign under T_R.*

Proof. Under T_R, each $a_\ell^m \mapsto (-1)^m a_\ell^{-m}$. Substituting into $\beta = \sum_m (\sum_{m_1+m_2=m} C\, a^{m_1} a^{m_2}) \overline{a^m}$, the three sign factors $(-1)^{m_1+m_2+m} = (-1)^{2m} = 1$ cancel. Reindexing $m \mapsto -m$ and applying the CG symmetry $C^{\ell,\,-m}_{\ell_1,\,-m_1;\,\ell_2,\,-m_2} = (-1)^{\ell_1+\ell_2-\ell}\, C^{\ell,\,m}_{\ell_1,\,m_1;\,\ell_2,\,m_2}$ yields (8) (with $(-1)^{\ell_1+\ell_2-\ell} = (-1)^{\ell_1+\ell_2+\ell}$).

For real signals, $\overline{a_\ell^m} = (-1)^m a_\ell^{-m}$. Taking the complex conjugate of β and re-indexing similarly yields (9). □

Lemma 13 (CG power is T_R-invariant)**.** *For all triples (ℓ_1, ℓ_2, ℓ): $P_{\ell_1,\ell_2,\ell}(T_R \cdot f) = P_{\ell_1,\ell_2,\ell}(f)$.*

Proof. Under T_R, the CG-projected component transforms as $(\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^M \mapsto (-1)^{\ell_1+\ell_2+\ell+M} (\mathbf{F}_{\ell_1} \otimes \mathbf{F}_{\ell_2})|_\ell^{-M}$. Taking the squared norm and re-indexing the sum over M leaves P unchanged. □

Proposition 14 (Vanishing of odd-parity entries with a repeated index)**.** *Let $\beta_{\ell_1,\ell_2,\ell}$ be the scalar bispectrum entry of a real signal f (so $\ell_1 + \ell_2 + \ell$ odd). If the index multiset $\{\ell_1, \ell_2, \ell\}$ contains a repeated value, then $\beta_{\ell_1,\ell_2,\ell}(f) \equiv 0$ as a polynomial in the SH coefficients of f.*

Proof. Define the symmetric scalar 3j contraction

$$T_{\ell_1,\ell_2,\ell_3}(f) := \sum_{m_1,m_2,m_3} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \end{pmatrix} a_{\ell_1}^{m_1} a_{\ell_2}^{m_2} a_{\ell_3}^{m_3}.$$

Converting the Wigner 3j into a Clebsch-Gordan coefficient via $\left(\begin{smallmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{smallmatrix}\right) = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} C^{j_3,\,-m_3}_{j_1,\,m_1;\,j_2,\,m_2}$ and applying the reality identity $a_{\ell_3}^{m_3} = (-1)^{m_3} \overline{a_{\ell_3}^{-m_3}}$ shows

$$\beta_{\ell_1,\ell_2,\ell}(f) = (-1)^{\ell_1+\ell_2} \sqrt{2\ell+1}\, T_{\ell_1,\ell_2,\ell}(f),$$

so β and T have the same vanishing locus. The 3j symbol is invariant under even column permutations and picks up $(-1)^{\ell_1+\ell_2+\ell_3}$ under odd ones. If two columns i, j satisfy $\ell_i = \ell_j$, the inserted vectors a_{ℓ_i}, a_{ℓ_j} are literally the same coefficient vector of f, so swapping those two columns leaves the contracted sum T unchanged while multiplying the 3j prefactor by $(-1)^{\ell_1+\ell_2+\ell_3}$. For odd parity this forces $T = -T$, hence $T \equiv 0$ and $\beta \equiv 0$. □

Remark 15 (Consequences for $\mathcal{T}_\ell$). Proposition 14 applies in particular to the family $(r, 2r-1, r)$: it has parity $4r-1$ (odd) and a repeated index, so $\beta_{r,2r-1,r} \equiv 0$ on real signals. Numerical evaluation at the seed-42 real witness confirms this: for $r = 2, \ldots, 7$ the entry is zero to double-precision rounding ($|\beta| \lesssim 10^{-14}$). Any formal definition of $\mathcal{T}_\ell$ must therefore exclude these triples (see (7)); we substitute a chain row with all-distinct indices in their place.

Corollary 16 (First nonzero odd-parity entry)**.** *The first bispectral entry with odd parity and generically nonzero imaginary part occurs at degree $\ell = 4$: $\beta_{2,3,4}$, with $\ell_1 + \ell_2 + \ell = 9$ (odd) and ℓ_1, ℓ_2, ℓ all distinct. At the deterministic witness (seed 42), $\mathrm{Im}(\beta_{2,3,4}) \approx 3.26 \neq 0$, and $\mathrm{Im}(\beta_{2,3,4})(T_R \cdot f) \approx -3.26$. Similarly, $\beta_{2,4,3}$ and $\beta_{3,4,2}$ are nonzero odd-parity entries at degree 4.*

Remark 17 (No parity-breaking at $\ell \leq 3$). For degrees ≤ 3, every odd-parity triple (ℓ_1, ℓ_2, ℓ) has at least two indices equal (since $\max(\ell_1, \ell_2, \ell) \leq 3$ and $\ell_1 + \ell_2 + \ell$ odd forces at least one pair to be equal in this range). By Proposition 14, all odd-parity entries vanish. Therefore, the T_R ambiguity cannot be resolved using only degrees ≤ 3.

Remark 18 (Why the augmented block is needed). The bootstrap block $\mathcal{T}_\ell$ alone is rank-deficient on the real-signal slice: at the seed-42 real witness, the $(2\ell+1) \times (2\ell+1)$ submatrix A_ℓ assembled from the $\mathcal{T}_\ell$ rows has SVD rank $\approx \ell+2$ rather than $2\ell+1$ for $\ell \in \{8, \ldots, 15\}$ (verified numerically; the deficit is generic over real witnesses). The mandatory even self-couplings and CG-power scalars are quadratic or cubic in $\mathbf{F}_\ell$ and supply the additional independent real equations needed to fill out the per-degree real Jacobian; the augmented per-degree block (linear $\mathcal{T}_\ell$ plus quadratic self-couplings plus CG-power scalars) is what Algorithm 2 assembles, and is what the implementation actually solves at degrees $\ell \geq 5$.

A.5 Comparison with existing results

A.6 Complexity in pixel space

On S^2, a band-limited signal at degree L is sampled on an equiangular grid of size $N_\theta \times N_\varphi \approx 2L \times 2L$, so $P = 4L^2$ and $L = \Theta(\sqrt{P})$.

Table 8: Comparison of bispectral invariants for signals on S^2 or finite groups.

Result	Invariant	Size	Complete?
Kakarala [Kakarala, 2012]	Full bispec.	$O(L^3)$	Yes (constr.)
Edidin & Satriano [Edidin and Satriano, 2024]	Full bispec. (band-lim.)	$O(L^3)$	Yes (generic)
Bendory et al. [Bendory et al., 2025]	Freq. marching	$O(L^2)$/shell	Yes (3 shells)
Mataigne et al. [Mataigne et al., 2024]	Sel. G-bispec. (finite)	$O(\|G\|)$	Yes (exact)
This work	**Sel.** $\mathrm{SO}(3)$ **on** S^2	$\boldsymbol{\Theta(L^2)}$	**Empirical (Section 6.2)**

Output size.

Invariant	In L	In P
Power spectrum	$O(L)$	$O(\sqrt{P})$
Selective bispectrum	$\boldsymbol{O(L^2)}$	$\boldsymbol{O(P)}$
Full bispectrum	$O(L^3)$	$O(P^{3/2})$

The selective bispectrum is *linear* in the number of pixels.

Computation cost.

Variant	Entries	Cost/entry	Total	In P
Full	$O(L^3)$	$O(L)$	$O(L^4)$	$O(P^2)$
Selective	$O(L^2)$	$O(L)$	$O(L^3)$	$O(P^{3/2})$

Concrete example. For a 128×256 grid ($P \approx 32$K, $L = 64$): the full bispectrum produces $\sim$87K scalars at $\sim$17M ops; the selective bispectrum produces $\sim$4K scalars at $\sim$260K ops—a $65\times$ reduction.

B Extended Benchmark Results

This appendix supplements the summary in Table 2 (main text) with scaling plots produced by `benchmarks/benchmark.py` on a single NVIDIA H100 80 GB GPU (PyTorch 2.10).

B.1 Coefficient-count scaling

B.2 Forward-pass timing

The selective forward pass is dominated by kernel-launch overhead rather than arithmetic, making it effectively free relative to backbone computation. The full bispectrum exhibits the expected super-linear growth and becomes impractical beyond $|G| \approx 256$ for C_n.

B.3 GPU throughput and inversion

All modules exhibit linear throughput scaling with batch size, confirming that the implementation is memory-bandwidth-bound rather than compute-bound at these scales.

C PCam: Extended Results

C.1 Published baselines

Table 9 places our results in the context of published methods on PCam and the closely related Camelyon16 benchmark. Our models operate at 68K–920K parameters and are trained on only 10% of the data.

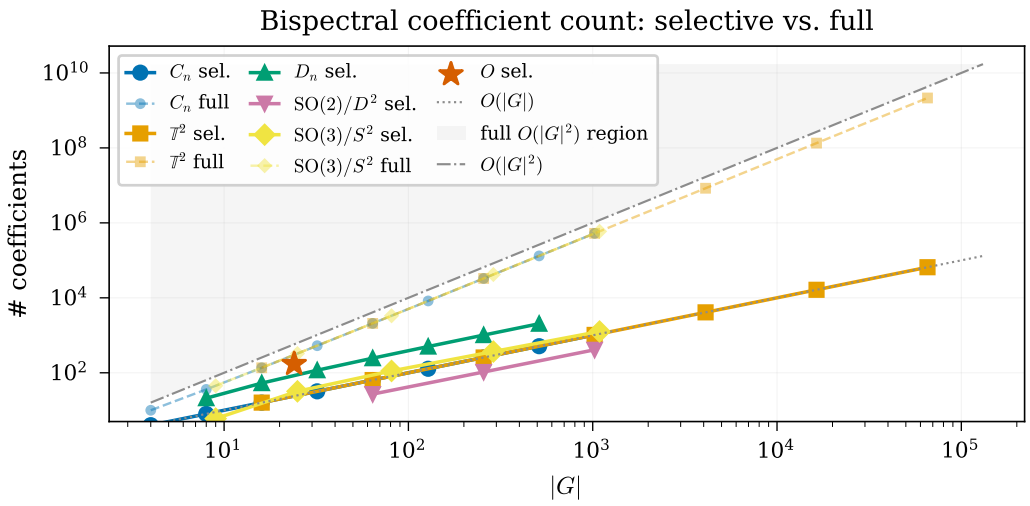

Figure 4: Bispectral coefficient count vs. group order $|G|$. Solid lines: selective ($O(|G|)$); dashed: full ($O(|G|^2)$). At $|G|$=1,024, selectivity yields a 512× reduction. Continuous groups (SO(2), SO(3)) are plotted at their band-limited signal dimension; full-bispectrum curves for D_n and the disk are omitted (backends not yet implemented).

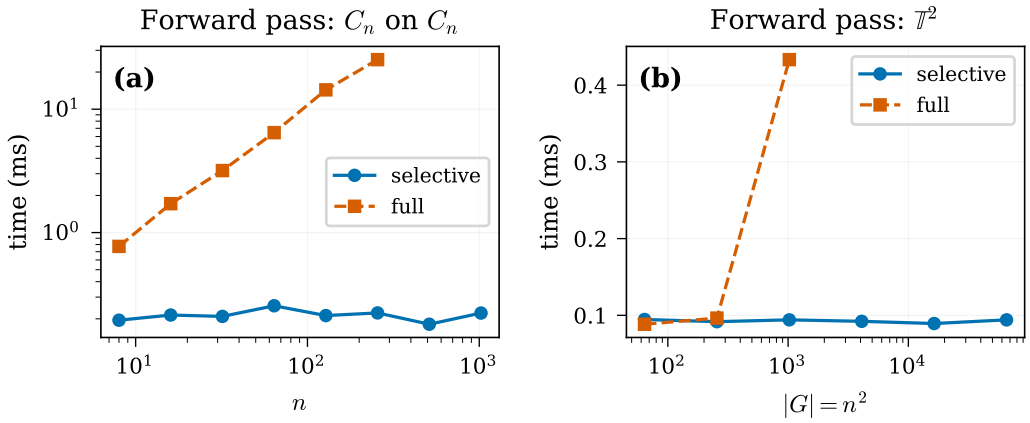

Figure 5: Forward-pass wall-clock time (batch = 16, H100 GPU). **(a)** C_n: the selective pass stays flat at ∼0.14 ms regardless of n, while the full bispectrum grows super-linearly (17 ms at n=256). **(b)** $\mathbb{T}^2$: selective is constant at ∼0.08 ms up to $|G|$=65,536.

C.2 Model variants

All equivariant models share a C_8-equivariant DenseNet [Huang et al., 2017] backbone with block configuration $(4, 4, 4)$, 24 initial channels, and compression factor 0.5, following the architecture of Veeling et al. [2018] but replacing the final invariant map. Six model variants are evaluated:

- **Standard**: vanilla CNN with random rotation/reflection augmentation and global average pooling. No group dimension, so ∼8× fewer parameters at matched growth rate.
- **NormReLU**: C_8-equivariant backbone with norm-based nonlinearity and max pooling over the group dimension.
- **Gated**: C_8-equivariant backbone with gated nonlinearity (doubles internal channels) and group max pool.

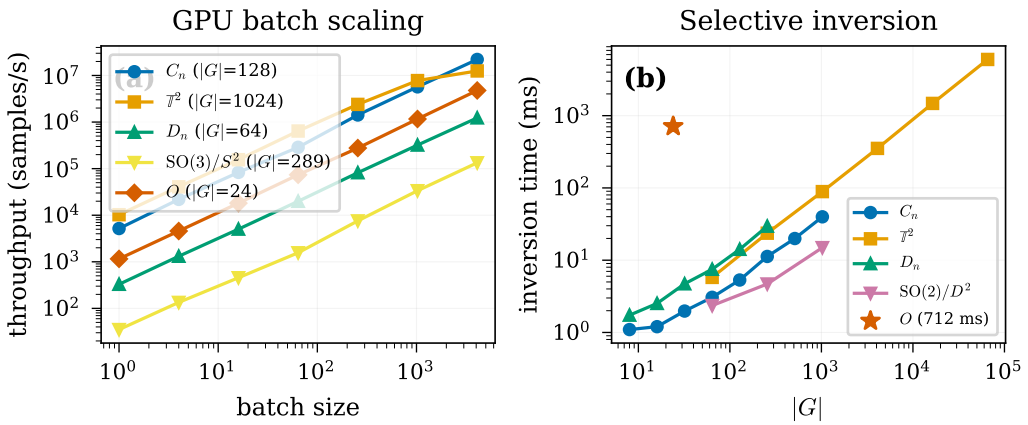

Figure 6: **(a)** GPU throughput (samples/s) vs. batch size for the selective forward pass. C_n and $\mathbb{T}^2$ exceed 10^7 samples/s; the octahedral module reaches 5.7×10^6 samples/s. All modules scale linearly with batch size. **(b)** Inversion wall-clock time. Abelian groups (C_n, D_n, SO(2)/disk) scale roughly linearly. SO(3) on S^2 inversion is not yet implemented.

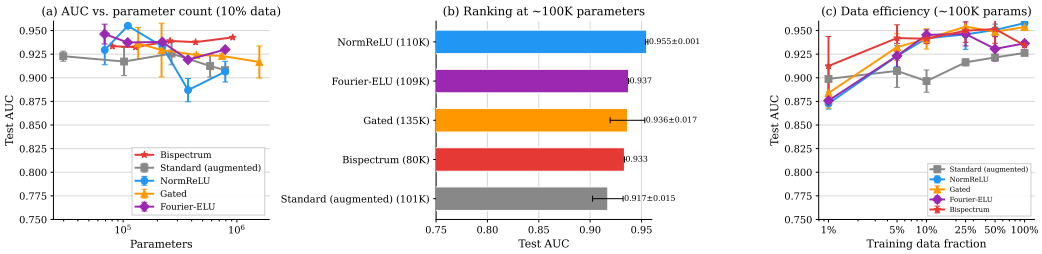

Figure 7: PCam Pareto analysis (`CnonCn` selective bispectrum, single seed, 10% training data). (a) Test AUC vs. parameter count across all five pooling variants. (b) Ranking at the ∼100K parameter budget. (c) Data efficiency at matched ∼100K parameters. Multi-seed statistics at matched capacity are in Table 3 (main text).

- **Fourier-ELU**: C_8-equivariant backbone with Fourier-space ELU nonlinearity and group max pool.
- **Bispectrum**: C_8-equivariant backbone with ReLU nonlinearity and selective G-bispectrum via `CnonCn`(n=8) as the terminal invariant map.

C.3 Training details

To enable a controlled Pareto comparison, we sweep the DenseNet growth rate $k \in \{3, 4, 6, 8, 12\}$ for equivariant models (and $k \in \{6, 12, 20, 30, 35\}$ for the standard CNN to cover a comparable parameter range), yielding models from ∼30K to ∼1.6M parameters. All models are trained on **10% of training data** (26,214 images) with AdamW [Loshchilov and Hutter, 2019] (lr $= 10^{-3}$, weight decay 10^{-4}), cosine warm restarts, mixed-precision training, and early stopping on validation AUC (patience 10, max 50 epochs).

C.4 Pareto analysis interpretation

Architectural stability (single seed, growth-rate sweep). The bispectrum's AUC varies by only ∼1pp across a 12× range of model sizes (80K–920K parameters), while norm swings by >7pp. For practitioners, bispectral pooling "just works" without growth rate tuning. The multi-seed matched-parameter results (Table 3) confirm this: bispectrum and Fourier-ELU have the lowest cross-seed variance (±0.004).

Table 9: Published baselines on PatchCamelyon from Veeling et al. [2018], trained on 100% data.

Method	Reference	Params	ACC	AUC
P4M-DenseNet (C_8)	Veeling et al. 2018	119K	0.898	0.963
P4M-DenseNet M	Veeling et al. 2018	19K	0.893	0.958
DenseNet (augmented)	Veeling et al. 2018	128K	0.881	0.951
DenseNet (no aug.)	Veeling et al. 2018	128K	0.876	0.955

Matched parameters. At matched parameters ($\sim$100K), all equivariant models outperform the standard CNN (AUC 0.896 ± 0.010, 3 seeds). Fourier-ELU leads slightly (0.945 ± 0.004), followed by NormReLU (0.942), bispectrum and gated (both 0.941). All equivariant methods cluster within $\sim$0.5pp of each other.

Scaling behavior. As parameters increase from the smallest to the largest configuration, standard, norm, and Fourier-ELU all degrade (by 1.8, 2.6, and 0.9 AUC points respectively), while the bispectrum improves (+0.9 points, reaching 0.943 at 920K params). This suggests that the bispectrum's complete invariant map scales more gracefully with capacity.

Data scaling (3 seeds). At 1% data the bispectrum leads all methods (0.912 ± 0.025), outperforming NormReLU (0.873), Fourier-ELU (0.876), and gate (0.884). By 10% all equivariant methods converge ($\sim$0.941–0.945). The bispectrum's data-efficiency advantage narrows as data increases, consistent with the hypothesis that complete invariants substitute for missing training examples in the low-data regime.

D OrganMNIST3D: Extended Results

D.1 Published baselines

Table 10 situates our results against published methods on the same benchmark. Our models are deliberately small ($\leq$463K parameters) to enable controlled ablation of the pooling mechanism. The relevant comparison is against equivariant methods at a similar scale.

Table 10: Published baselines on OrganMNIST3D for context. Accuracy differences with respect to large-scale models reflect the deliberate parameter constraint of our ablation study.

Method	Reference	Params	ACC	AUC
ResNet-18 + 3D	Yang et al. 2023	33M	0.907	0.996
ResNet-18 + ACS	Yang et al. 2023	11M	0.900	0.994
ILPOResNet-50	Zhemchuzhnikov and Grudinin [2024]	38K	0.879	0.992
EquiLoPO (local train.)	Zhemchuzhnikov and Grudinin [2025]	418K	0.866	0.991
SE3MovFrNet	Sangalli et al. [2023]	—	0.745	—
Regular SE(3) conv	Kuipers and Bekkers [2023]	172K	0.698	—

D.2 Rotation analysis

Figure 8 shows that all equivariant models achieve identical performance on the original and octahedrally rotated test sets ($\sigma_{\mathrm{rot}} \approx 0$), while the augmentation-trained standard CNN degrades under rotation.

D.3 Wider channels

To test whether the advantage of bispectral completeness persists at higher capacity, we run max pool and bispectrum at channel widths $(8, 16)$ and $(16, 32)$ with full training data.

At $(8, 16)$ the two methods are equivalent ($74.5 \pm 0.6\%$ bispectrum vs. $74.3 \pm 1.5\%$ max pool, both at $\sim$1.5–1.7M parameters). At $(16, 32)$ max pool reaches $78.5 \pm 3.2\%$ while bispectrum drops to $68.5 \pm 3.9\%$—the 6.3M parameter bispectrum model likely overfits the 971 training samples. This suggests that the bispectrum's completeness advantage is most pronounced in the data-limited rather than the parameter-limited regime, consistent with the data efficiency findings in the main text.

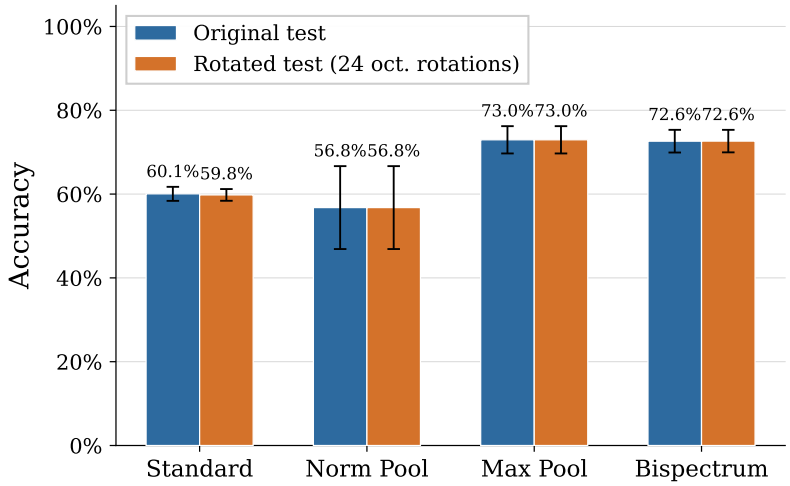

Figure 8: OrganMNIST3D test accuracy on the original and octahedrally rotated test sets. Error bars show standard deviation over three seeds. All equivariant models achieve identical performance under rotation ($\sigma_{\text{rot}} \approx 0$), while the augmentation-trained standard CNN degrades.

Table 11: OrganMNIST3D accuracy at wider channel configurations (mean $\pm$ std over 3 seeds, full training data).

Channels	Params	Max Pool	Bispectrum
$(4, 8)$	$\sim$375K / 463K	$.730 \pm .033$	$.726 \pm .027$
$(8, 16)$	$\sim$1.5M / 1.7M	$.743 \pm .015$	$.745 \pm .006$
$(16, 32)$	$\sim$6.0M / 6.3M	$\mathbf{.785 \pm .032}$	$.685 \pm .039$

E Experimental Details

E.1 Hyperparameters

Table 12 summarises the training hyperparameters for all three experimental settings.

Table 12: Training hyperparameters across experiments.

	PCam	OrganMNIST3D	Sph. MNIST
Optimizer	AdamW	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}	10^{-4}
Batch size	64–256	16–64	256
Scheduler	Cosine warm restarts	—	—
Max epochs	50	100	50
Early stopping	Val AUC, pat. 10	Val AUC, pat. 15	Val ACC, pat. 10
Seeds	42, 123, 456†	42, 123, 456	42, 123, 456
Mixed precision	Yes	No	No

†Pareto figure uses seed 42; matched-parameter table and data efficiency use seeds 42, 123, 456.

E.2 Hardware

All experiments were run on a single NVIDIA H100 GPU (80 GB). Training wall-clock times range from $\sim$5 minutes (OrganMNIST3D, 971 samples) to $\sim$20 minutes (PCam, 26K samples at 10% data). Spherical MNIST bispectrum computation (forward pass through `SO3onS2` at $\ell_{\max} = 15$) takes $\sim$0.3s per batch of 64 on GPU.

E.3 Data preprocessing

PCam. PatchCamelyon [Veeling et al., 2018] consists of 327,680 RGB patches of size 96×96 pixels extracted from whole-slide images of lymph node sections [Ehteshami Bejnordi et al., 2017]. We use the standard train/val/test split. Images are loaded from HDF5 files and normalised to $[0, 1]$. The standard (non-equivariant) baseline uses random rotation and reflection augmentation during training; all equivariant models are trained without augmentation.

OrganMNIST3D. OrganMNIST3D [Yang et al., 2023] contains 1,742 abdominal CT organ volumes of size $28 \times 28 \times 28$ voxels with 11 organ classes (train/val/test = 971/161/610). Volumes are normalised to $[0, 1]$. The standard baseline uses random octahedral augmentation (one of 24 rotations per sample); equivariant models are trained without augmentation. Kernel rotations are implemented as exact voxel permutations on the 3^3 grid, avoiding interpolation artifacts.

Spherical MNIST. MNIST digits are projected onto the sphere via stereographic projection following Cohen et al. [2018] and sampled on a 64×128 equiangular grid. The selective bispectrum is computed at $\ell_{\max} = 15$, producing 384 complex invariant coefficients. The model input is obtained by applying $\mathrm{sign}(x)\log(1 + |x|)$ separately to the real and imaginary parts of each coefficient (a sign-preserving, per-channel injective transform), giving 768 real features. These features are fed to a 3-layer MLP: Linear(768, 256)–BN–ReLU–Linear(256, 128)–BN–ReLU–Linear(128, 10). The power spectrum MLP uses the same structure with hidden=128: Linear(16, 128)–BN–ReLU–Linear(128, 64)–BN–ReLU–Linear(64, 10). The standard CNN has 4 convolutional layers (16–32–64–64 channels) followed by adaptive average pooling and a Linear(1024, 128)–ReLU–Linear(128, 10) classifier.

E.4 Aggregation procedures

PCam. The Pareto figure (Figure 7) uses a single seed (42) across the full growth-rate sweep. The matched-parameter table (Table 3) and data efficiency analysis use 3 seeds (42, 123, 456) per configuration.

OrganMNIST3D. All results in Table 4 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is the standard deviation of test accuracy across all 24 octahedral rotations of the test set. The data efficiency analysis (Figure 1 in the main text) uses 3 seeds per data fraction.

Spherical MNIST. All results in Table 5 report mean $\pm$ std over 3 seeds (42, 123, 456). Rotation robustness (σ_{rot}) is measured by applying 10 random $\mathrm{SO}(3)$ rotations to the entire test set and computing the standard deviation of accuracy, yielding 30 measurements per model (10 rotations $\times$ 3 seeds).

E.5 Spherical MNIST: bispectrum reconstruction protocol

This section gives the protocol behind Figure 3 (Section 6.2). The implementation lives in `paper/experiments/spherical_mnist_reconstruction/reconstruct.py`.

Setup. Spherical MNIST is rendered onto an equiangular $64{\times}128$ lat–lon grid using the same stereographic projection as the classifier, then band-limited at $L = 12$ via a forward/inverse SHT roundtrip (`torch_harmonics.RealSHT`). For each digit we draw a target rotation $R \in \mathrm{SO}(3)$ uniformly at random and form $f := R \cdot f_0$. We compute the SO(3)-on-S^2 selective bispectrum $\beta := \Phi_{\mathrm{sel}}(f)$.

Reconstruction. Starting from a Gaussian initialization $\hat{f}_0 \sim \mathcal{N}(0, I)$ on the $64{\times}128$ grid, we optimize $\hat{f}$ with Adam (initial LR 5×10^{-2}, cosine annealing to 5×10^{-4}, 8000 steps) to minimize the relative complex L2 loss $\|\beta(\hat{f}) - \beta\|^2/\|\beta\|^2$. After every step we project $\hat{f}$ back to the band-limited subspace via $\mathrm{SHT}^{-1} \circ \mathrm{SHT}$ truncated at L to suppress the unconstrained null space. We run 4 random restarts and keep the $\hat{f}$ with the lowest bispectrum residual. Reconstruction typically reaches $\|\beta(\hat{f}) - \beta\|/\|\beta\| \sim 10^{-3}$, near the SHT-discretization invariance floor for this grid.

$\mathrm{SO}(3)$ **alignment.** The bispectrum is invariant on $\mathrm{SO}(3)$-orbits, so $\hat{f}$ lives somewhere in the orbit of f rather than at f itself. We resolve the orbit ambiguity by Procrustes-style fitting: we parameterize $\hat{R} \in \mathrm{SO}(3)$ as a unit quaternion (auto-normalized) to avoid gimbal lock, and minimize $\|\hat{R} \cdot \hat{f} - f\|^2$ with Adam (12 random restarts, 200 steps each, cosine-annealed LR). Rotations of the spherical image use bilinear interpolation in lat–lon coordinates (`bispectrum.rotate_spherical_function`).

Numerical floor. On a 64×128 grid, two SHT roundtrips of two $\mathrm{SO}(3)$-rotated copies of the same signal already differ by $\|f' - f\|/\|f\| \approx 2 \times 10^{-2}$ in image space due to bilinear-interpolation discretization; this is the noise floor below which alignment cannot improve. The aligned residuals reported in Figure 3 (0.1–0.3) sit one to two orders of magnitude above this floor and are dominated by gradient descent local optima on the bispectrum reconstruction step, not by missing information.

E.6 Dataset licenses

- **PCam**: CC0 1.0 Universal (public domain).
- **OrganMNIST3D**: CC BY 4.0.
- **Spherical MNIST**: derived from MNIST, which is publicly available for research use.

F API Example

```
import torch
from bispectrum import CnonCn, SO2onDisk, OctaonOcta

bsp = CnonCn(n=8, selective=True)
f = torch.randn(16, 8)
beta = bsp(f)
f_rec = bsp.invert(beta)
```

NeurIPS Paper Checklist

1. **Claims**
   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: [Yes]
   Justification: The abstract and introduction list four contributions (selective bispectrum theory, completeness proof, `nn.Module` library, and experiments on three benchmarks) that are substantiated in Sections 3–4 and Appendices A–B.
   Guidelines:
   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.
2. **Limitations**
   Question: Does the paper discuss the limitations of the work performed by the authors?
   Answer: [Yes]
   Justification: Section 6 contains a dedicated Limitations paragraph discussing the genericity assumption, bootstrap verification scope ($L \leq 100$), octahedral inversion recovery rate, and missing experimental comparisons. Appendix A includes a "Limitations and open questions" subsection.
   Guidelines:
   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.
3. **Theory assumptions and proofs**
   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   Answer: [Yes]

Justification: Theorem 1 states all assumptions (genericity, gauge-fixing) and a proof sketch appears in Section 3.4. The complete proof with 10 lemmas and 6 propositions is in Appendix A. The octahedral inversion gap analysis is in Appendix B.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: All hyperparameters are in Appendix F (Table 12), architectures in Appendices D–E, optimizer and scheduler details in Table 12, and random seeds are specified. The code is provided as an anonymous repository.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?
Answer: [Yes]
Justification: An anonymous code repository is provided with the submission. All three datasets (PCam, OrganMNIST3D, Spherical MNIST) are publicly available. Sweep scripts and training code are included.
Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**
Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?
Answer: [Yes]
Justification: Sections 4.1–4.3 describe experimental setups. Full hyperparameters are in Table 12 (Appendix F), architecture details in Appendices D–E, data preprocessing and splits in Appendix F.
Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**
Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
Answer: [Yes]
Justification: Tables 4–5 (OrganMNIST3D, Spherical MNIST) report mean $\pm$ standard deviation over 3 seeds. Table 3 (PCam) uses a single seed for the Pareto sweep, which is noted in Appendix F. The aggregation procedure is described in Appendix F.
Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).

- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**
Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?
Answer: [Yes]
Justification: Appendix F states that all experiments were run on a single NVIDIA H100 GPU (80 GB). Wall-clock times are available in the code repository.
Guidelines:
   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**
Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?
Answer: [Yes]
Justification: The work develops mathematical invariants and a software library evaluated on publicly available datasets. No human subjects, private data, or dual-use concerns are involved.
Guidelines:
   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**
Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
Answer: [N/A]
Justification: This is foundational mathematical and software research on group-theoretic invariants. We do not foresee direct negative societal impacts.
Guidelines:
   - The answer [N/A] means that there is no societal impact of the work performed.
   - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
   - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
   - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to

generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.

- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released asset is a deterministic mathematical library for computing group-theoretic invariants. It poses no risk for misuse.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All datasets are cited (PCam: CC0 1.0, OrganMNIST3D: CC BY 4.0, MNIST: public domain for research). Dataset licenses are listed in Appendix F.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The library is released under the MIT license with documentation, API reference, and usage examples. Section 3 describes the design and interface.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**
Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?
Answer: [N/A]
Justification: This work does not involve crowdsourcing or human subjects.
Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**
Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?
Answer: [N/A]
Justification: This work does not involve human subjects.
Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**
Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.
Answer: [N/A]
Justification: LLMs are not part of the core methodology of this work.
Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

Algorithm 2 Augmented selective index set for the $\mathrm{SO}(3)$ bispectrum on S^2 at band-limit L. Returns bispectral triples $\mathcal{S}_\beta$ and CG power triples $\mathcal{S}_P$. This extends the BFS-based selective index construction of Mataigne et al. [2024] from finite groups to $\mathrm{SO}(3)$ on S^2, adding the CG power augmentation step that is specific to real signals.

Input: Band-limit L
Output: Bispectral set $\mathcal{S}_\beta$, CG power set $\mathcal{S}_P$
$\mathcal{S}_\beta \leftarrow \{(0,0,0)\}$; $\mathcal{S}_P \leftarrow \emptyset$
Seed block: add to $\mathcal{S}_\beta$ all triples (ℓ_1, ℓ_2, ℓ) with $0 \le \ell_1 \le \ell_2 \le L_0 = 4$ and $|\ell_1 - \ell_2| \le \ell \le \min(\ell_1 + \ell_2, L_0)$ {constant-size full-bispectrum seed}
for $\ell = 1, \ldots, L$ **do**
  budget $\leftarrow 2\ell + 1$
  if $\ell = 4$ **then**
    budget $\leftarrow 10$ {keep all chain+cross at $\ell = 4$}
  end if
  candidates $\leftarrow [\,]$
  if $4 \le \ell \le 7$ **then**
    append the explicit low-degree linear block $\mathcal{T}_\ell^{\mathrm{small}}$ to candidates
  else if $\ell \ge 8$ **then**
    for $a = 1, \ldots, \ell{-}1$ **do**
      append $(a,\ \ell,\ \ell{-}a)$ to candidates
    end for
    for $a = 2, \ldots, \ell{-}1$ **do**
      if not (ℓ is odd **and** $a = (\ell{+}1)/2$) **then**
        append $(a,\ \ell,\ \ell{-}a{+}1)$ to candidates {skip $(r, 2r{-}1, r) \equiv 0$ on real signals, Prop. 14}
      end if
    end for
    for $a = 1, \ldots, 4$ **do**
      append $(a,\ \ell{-}a,\ \ell)$ to candidates
    end for
    if ℓ is odd **then**
      append $(2,\ \ell{-}1,\ \ell)$ to candidates {compensate excluded $(r, 2r{-}1, r)$ row}
    end if
  else
    append low-degree chain candidates
    if $\ell \ne 2$ **then**
      append cross candidates {$\beta_{1,2,1}$ excluded}
    end if
  end if
  append $(0,\ \ell,\ \ell)$ to candidates {power entry}
  if $\ell \notin \{1,\ 2\}$ **then**
    for $\ell' = \ell{-}1, \ldots, 0$ **do**
      append $(\ell,\ \ell,\ \ell')$ to candidates
    end for
  end if
  if $\ell = 2$ **then**
    append $(\ell,\ \ell,\ \ell)$ to candidates $\{\beta_{2,2,2}\}$
  end if
  Remove entries that vanish for real signals; deduplicate against $\mathcal{S}_\beta$ already added by the seed block
  $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup$ first $\min(\text{budget},\ |\text{candidates}|)$ live entries
  {**Mandatory even self-couplings**}
  for $\ell' = 2, 4, \ldots, \lfloor \ell/2 \rfloor \cdot 2$ with $\ell' \le \ell$ and $\ell \ge 3$ **do**
    $\mathcal{S}_\beta \leftarrow \mathcal{S}_\beta \cup \{(\ell,\ \ell,\ \ell')\}$
  end for
  {**CG power augmentation**}
  Greedily add $P_{\ell_1,\ell_2,\ell'}$ (with $\max(\ell_1, \ell_2) = \ell$, $|\ell_1{-}\ell_2| \le \ell' \le \ell_1{+}\ell_2$) to $\mathcal{S}_P$ until the Jacobian rank of the per-degree restriction $\Phi_{\mathrm{sel}}|_\ell$ matches the number of unknowns
end for
return $(\mathcal{S}_\beta,\ \mathcal{S}_P)$